BLOWING UP COMMUTES WITH FLAT BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The statement carries the author's slogan environment 'Blowing up commutes with flat base change'; it is part of the counted statement span and is preserved verbatim, and the chapter's retained preamble declares the environment (with the verbatim package) so the comment body typesets as intended.

Editorial note. The statement contains an xymatrix square; it is preserved verbatim with the chapter's own Xy-pic setup.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The ideal sheaf of an inverse image closed subscheme. Schemes, Definition schemes-definition-inverse-image-closed-subscheme, used at proof lines 8369-8372. Identifies I_1 as the image of g^*I_2.
- Base change of a relative Proj. Constructions, Lemma constructions-lemma-relative-proj-base-change, used at proof lines 8372-8375. Describes X_1 x_{X_2} X'_2 as a relative Proj of the pulled back graded algebra.

Source reference. Original lemma 01O3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2180–2180. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 01JV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2028–2028. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $X_1 \to X_2$ be a flat morphism of schemes. Let $Z_2 \subset X_2$ be a closed subscheme. Let Z_1 be the inverse image of Z_2 in X_1. Let X_i' be the blowup of Z_i in X_i. Then there exists a cartesian diagram*

$$\begin{array}{ccc} X_1' & \longrightarrow & X_2' \\ \downarrow & & \downarrow \\ X_1 & \longrightarrow & X_2 \end{array}$$

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of schemes.

Proof. Let $\mathcal{I}_2$ be the ideal sheaf of Z_2 in X_2. Denote $g : X_1 \to X_2$ the given morphism. Then the ideal sheaf $\mathcal{I}_1$ of Z_1 is the image of $g^*\mathcal{I}_2 \to \mathcal{O}_{X_1}$ (by definition of the inverse image, see Schemes, Definition 01JV). By Constructions, Lemma 01O3 we see that $X_1 \times_{X_2} X_2'$ is the relative Proj of $\bigoplus_{n\geq 0} g^*\mathcal{I}_2^n$. Because g is flat the map $g^*\mathcal{I}_2^n \to \mathcal{O}_{X_1}$ is injective with image $\mathcal{I}_1^n$. Thus we see that $X_1 \times_{X_2} X_2' = X_1'$. $\square$